



Introduction to OS

Rab Nawaz Jadoon

Assistant Professor

COMSATS IIT, Lahore
Pakistan

Department of Computer Science

DCS

COMSATS Institute of
Information Technology

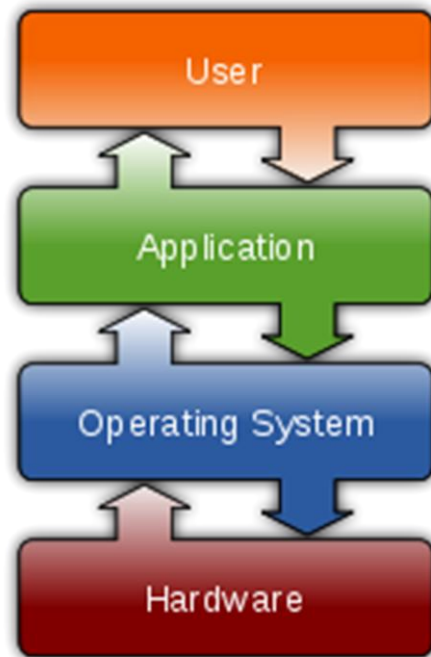
Operating System Concepts

Operating System

■ Operating System

- It is a **system software** that run the hardware. OR
- Group of programs or routines which acts as an interface between user and computer. OR
- Collection of programs implemented in either software or firmware that control the hardware resources of the computer system. OR
- Manage and control all the resources and events happened in the computer system. OR
- It is an **animated spirit** of the computer.

Operating System



Operating System Placement



Operating System functioning

Needs of Operating System

- It provides high level of interface between user and machine.
- Also provides the efficient use of the hardware.
- The overall objective is to achieve the overall efficiency in processing.
- The OS is the first program loaded into the computer memory, when PC is switched ON.
 - Other portion swapped in and out upon usage.

Boot sequence for the IBM PC

■ Booting

- The process of loading programs from Secondary storage media to RAM.
- Initial startup of the computer.
- Boot Loader is a program used to load the OS.

■ Process

- Upon turn ON, some count down on the screen, peripheral devices are detected.
- If Hard Disk/Floppy Disk is not there then
 - "No boot device" msg appear on the screen
 - If HDD is present but it is not system formatted then "Insert System Disk" msg appear on the screen.

Boot sequence for the IBM PC

- After some time peripheral devices lights are ON

■ Question???

- If we have no HDD and FDD then from where these messages appear on the screen????? Like no boot device etc has been displayed.
- This mean that there is some storage which contains these messages.
 - The answer to all above questions is YES that there is a CHIP on the motherboard, called ROM BIOS.
 - ROM/BIOS is an IC having programs known as Firmware.
 - Hardware component containing S/W called Firmware.

Boot sequence for the IBM PC

- ROM/BIOS contains firmware responsible to initiate the function of the computer.
 - This firmware → IPL (Initial Program Loader)
 - IPL is the first program which is executed whenever computer is turned ON.
 - ROM/BIOS → first chip which is turned ON.
 - IPL checks the presence of the storage device.
 - Then to go to search OS on it.
 - If no OS then "No boot device available" msg displayed by IPL.

Boot sequence for the IBM PC

- If storage device present then
 - IPL executes the boot strap routines (stored on the boot sector of HDD)
 - BOOT Sector → first sector of the first track
 - Boot Strap Routines → Load the OS into the computer memory.
 - Finds the OS kernel on the HDD, load into memory and jumps to the initial address to begin the OS's execution.

Available operating systems in the market

■ Popular modern operating system includes,

- Android
- BSD (Berkeley Software Distribution)
- iOS
- Linux
- MAC OS X
- Microsoft Windows
- Window phone
- IBM z/OS
- Symbian etc.
 - All these except Windows and z/OS, share roots in UNIX

Modern Operating Systems



- Founded in Palo Alto, California, US in Oct 2003
- It is a LINUX based OS for mobile devices such as smart phones, tablet computers etc.
 - Android has a large community of developers writing applications ("apps") that extend the functionality of the devices. Developers write primarily in a customized version of Java.
 - In Oct 2012, there were more than **700,000** apps available for Android.
 - Estimated number of applications downloaded from Google Play was **25 billion**.

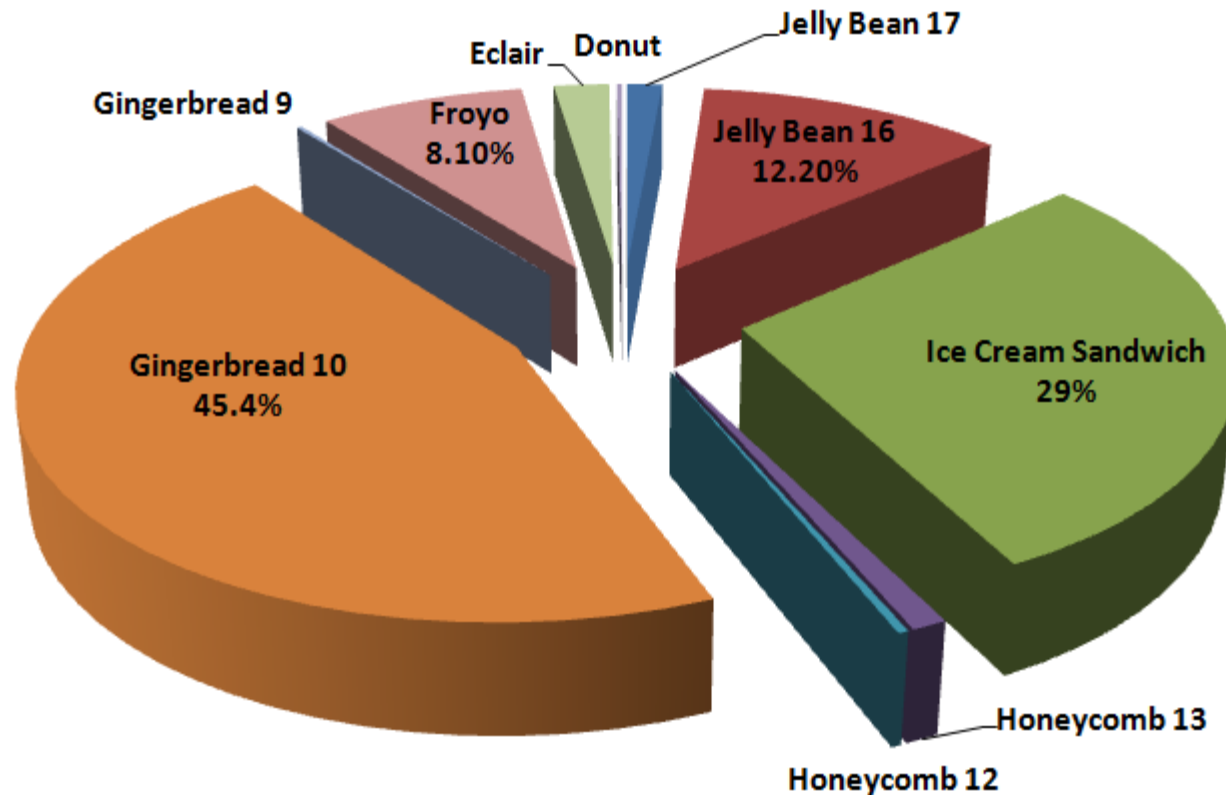
- Android became the world's leading smartphone platform at the end of 2010.
 - For the third quarter of 2012, Android had a 75% smartphone market share worldwide.
 - At the half of 2012, there were 500 million devices activated and 1.3 million activations per day.
 - Alphabetical codename of android are,
 - Cupcake, Donut, Eclair, Froyo, Gingerbread, Honeycomb, Ice Cream Sandwich and Jelly Bean.



- As of 2013, over **500 million active** devices use the Android OS worldwide.
 - The most recent major Android update was Jelly Bean 4.2, which was released on commercial devices in Nov 2012.
 - The first devices to run Android 4.2 were **LG's Nexus 4 and Samsung's Nexus 10**, which were released on 13 November 2012.

Android (as per Jan 2013)

Android version usage percentage



[http://en.wikipedia.org/wiki/Android_\(operating_system\)](http://en.wikipedia.org/wiki/Android_(operating_system))

■ Berkeley Software Distribution

- Berkeley Software Distribution (BSD, sometimes called Berkeley Unix) is a Unix operating system.
- It is developed and distributed by the Computer Systems Research Group (CSRG) of the University of California, Berkeley, from 1977 to 1995.



- iOS (previously iPhone OS) is a mobile operating system developed and distributed by Apple Inc.
 - Originally released in 2007 for the iPhone and iPod Touch, it has been extended to support other Apple devices such as the iPad and Apple TV.
 - Unlike Microsoft's Windows CE (Windows Phone) and Google's Android, Apple does not license iOS for installation on non-Apple hardware.
 - As of June 12, 2012, Apple's App Store contained more than 650,000 iOS applications.
 - It had a 23% share of the smartphone operating system units sold in the first quarter of 2012.

- Unix-like computer operating system assembled under the model of free and open source software development and distribution.
 - The defining component of Linux is the Linux kernel, an operating system kernel first released on 5 Oct 1991 by Linus Torvalds.
 - First developed for Intelx86 based personal computers.
 - Leading OS on servers and other big supercomputers run some variant of Linux, Including the 10 fastest machines of the world.
 - Popular distributions include, ubuntu, fedora and openSUSE etc.



- **Mac OS X**, is a series of Unix based GUI operating systems developed, marketed, and sold by Apple Inc.
 - It exclusively runs on Macintosh computers since 2002.
 - The latest version MAC OS X 10.7 “Lion” was the first version of OS X which supported 32 bit Intel processors and run exclusively on 64bit Intel CPUs.
 - Important version includes,
 - Tiger, Leopard, Snow Leopard, Lion etc.



Microsoft Windows

- Microsoft Windows is a series of GUI OS developed, marketed, and sold by Microsoft.
 - Microsoft introduced Windows on November 20, 1985 as an add-on to MS-DOS in response to the growing interest in graphical user interfaces (GUIs).
 - It dominate the world's personal computer market with over 90% market share.
 - The most recent client version of Windows is Windows 8.
 - The most recent server version is Windows Server 2008 R2.
 - The most recent mobile version is Windows Phone 7.5



Windows Phone

- It is a Mobile OS developed by Microsoft.
- It was launched in the second half of 2010.
- Specifically for PDAs and other handheld gadgets.

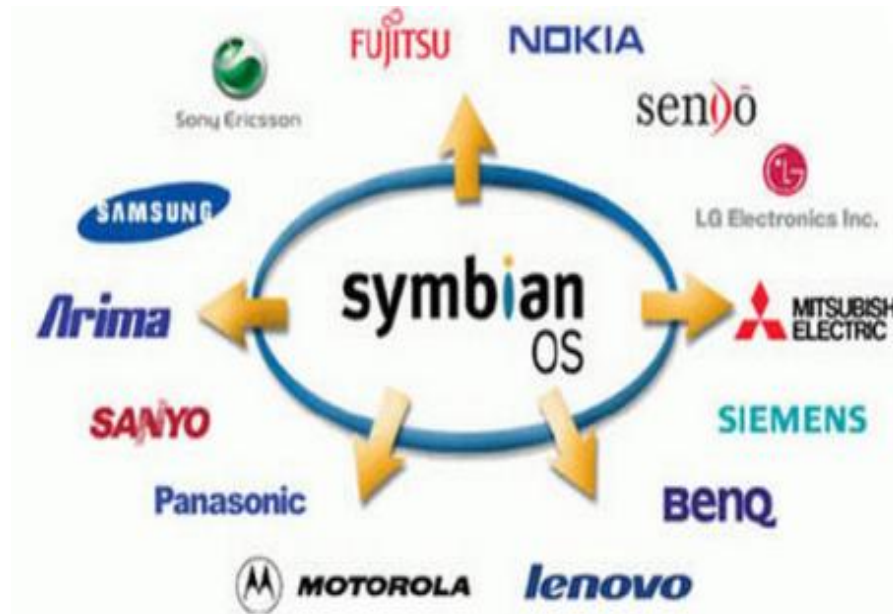


Windows Phone

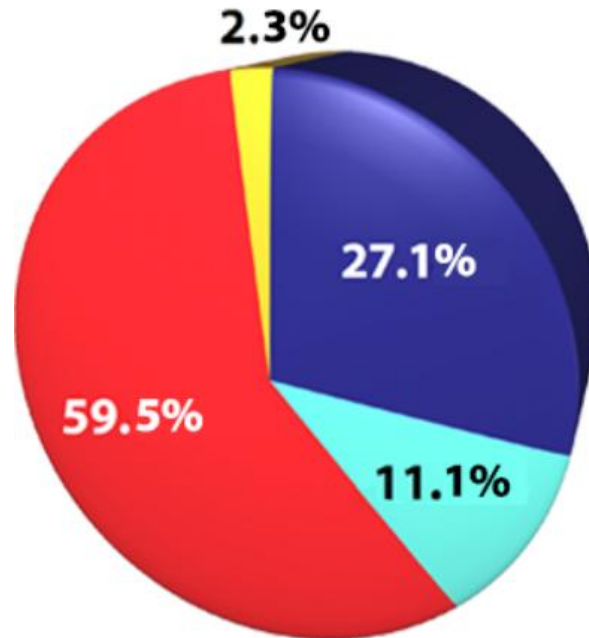
- **z/OS** is a 64-bit operating system for mainframe computers, produced by IBM.
- **z/OS** was first introduced in October 2000.
- **z/OS** offers the attributes of modern operating systems.
- Backward compatibility is one of **z/OS's** central design philosophies.
- IBM's **z/OS** has batch processing capabilities



- **Symbian** is a mobile operating system (OS) and computing platform.
- **Symbian** is designed for smart phones and currently maintained by Accenture.



OS Comparisons



According to Dec 2012

- Windows XP
- Windows Vista
- Windows 7
- Windows 8

Types of OS

- Within the broad family of operating systems, there are generally four types,
 - categorized based on the types of computers they control and the sort of applications they support. the categories are,
 - Real time Operating System (RTOS)
 - Single User, single task OS
 - Single User, Multitasking OS
 - Multiuser OS
 - Distributed OS
 - Embedded OS
 - Mobile OS
 - Batch OS etc.

- RTOS are used to control machinery, scientific instruments and industrial systems.
 - It is a multitasking operating system.
 - The main object of real-time operating systems is their quick and predictable response to events.
 - An RTOS typically has very little user-interface capability, and no end-user utilities, since the system will be a "sealed box" when delivered for use.

Single User, Single Task

- As the name implies, this operating system is designed to manage the computer so that one user can effectively do one thing at a time.
 - The Palm OS for Palm handheld computers is a good example of a modern single-user, single-task operating system.
 - E.g. PDA, and MS-DOS
 - Further extended to support smart phones etc.

Single User, Multitask

- This is the type of operating system most people use on their desktop and laptop computers today.
 - Microsoft's Windows and Apple's Mac OS platforms are both examples of operating systems that will let a single user have several programs in operation at the same time.

- A multi-user operating system allows many different users to use computer's resources simultaneously.
 - various users should be balanced, and that each of the programs they are using/executing has sufficient and separate resources.
 - For example Unix, VMS and mainframe operating systems, such as MVS, are examples of multi-user operating systems.

- An operating system that manages a group of independent computers and makes them appear to be a single computer is known as a distributed operating system.
 - The development of networked computers that could be linked and made to communicate with each other, gave rise to distributed computing.
 - Distributed computations are carried out on more than one machine. When computers in a group work in cooperation, they make a distributed system.

Embedded system

- They are designed to operate on small machines like PDAs with less autonomy.
 - They are able to operate with a limited number of resources.
 - They are very compact and extremely efficient by design.
 - Windows CE, FreeBSD and Minix 3 are some examples of embedded operating systems.

- A mobile OS controls a mobile device and its design supports wireless communication and mobile applications.
 - It has built-in support for mobile multimedia formats. Tablet PCs and smart phones run on mobile operating systems.

Batch Processing Systems

- In batch processing systems, programs are collected, grouped and processed on a later date.
 - There is no prompting the user for inputs as input data are collected in advance for future processing.
 - Input data are collected and processed in batches, hence the name batch processing.
 - IBM's z/OS has batch processing capabilities.

Online and Offline Processing

- In **online processing** of data, processes are executed under the control of the computer's central processing unit.
- When processes are not executed under direct control of the CPU, the processing is referred to as **offline**.
 - E.g. batch processing. Here, the batching or grouping of data can be done without user and CPU intervention; it can be done **offline**.
 - But the actual process execution may happen under direct control of the processor, **that is online**.

Important terminologies

■ Task

- When a program is requested for execution it is called a task.
- Before pressing enter key.

■ Job

- When the OS checks the validity of the task by searching it in internal page table, if present then it is said to be a valid job for further processing.

■ Process

- The job is then loaded into computer memory, after memory allocation, the program is set into execution.

Important terminologies

- When we write on the console,
 - **C:\>dir**
 - A task has been given to OS
 - OS checks its validity by searching it into internal page table, if found, then it is valid, become a job
 - When it become a job, OS must load it into memory.
 - Now this program under execution along with its environment is called a **Process**.

Important terminologies

- There are three processing systems,
 - Multiprogramming
 - Multiprocessing
 - Multitasking
- **Multiprogramming**
 - There is only one processor
 - More than one user (terminals)
 - It is the function of OS to share CPU time among all the users equally.

Important terminologies

■ Multiprocessing

- Multiprocessing is the use of two or more central processing units (CPUs) within a single computer system.
- In a multiprocessing system, all CPUs may be equal, or some may be reserved for special purposes.
 - Kernel-mode code may be restricted to only one processor.
 - Whereas user-mode code may be executed in any combination of processors.
 - Multiprocessing systems are often easier to design if such restrictions are imposed, but they tend to be less efficient than systems in which all CPUs are utilized.

Important terminologies

- Systems that treat all CPUs equally are called **symmetric multiprocessing (SMP) systems**.
 - Has UMA (Uniform Memory Access)
- In systems where all CPUs are not equal, system resources may be divided in a number of ways, including **asymmetric multiprocessing (ASMP)**, non-uniform memory access (NUMA i.e. local memory) multiprocessing, and clustered multiprocessing.

Important terminologies

■ Multitasking Systems

- There is only one processor and one user.
- User can activate more than one program/task at a time.
- In these system it is the responsibility of the OS to schedule the multiple jobs for their fair share processing.



Thanks